Supplementary material

Reproducible Sparse-Concept and Calibration Diagnostics for Knowledge Tracing

IJIET review copy. Not a classroom trial. ASSISTments locked cells unchanged.

These tables accompany the main manuscript (Tables S1–S10 and the cluster-aware regression). Table S1 is the full Brier/UNC/REL/RES grid; the main-text Brier extract is Table 6. Table S2 is the complete within-KC controlled-sparsification grid (the 12-row main-text extract was removed; the main paper keeps one pointer sentence). Tables S3–S6 are the simulated τ probe, not contribution (ii): S3 τ grid, S4 occupancy policies, S5 seed-42 FAR (locked 0.196/0.268), S6 four-partition ΔFAR. Table S7 is the three-cut frequency-grid check on frozen T-KT and DKT CSVs. Table S8 is stratum-level AUC/ACC. Table S9 records the baseline inventory versus the P0 roadmap (BKT not scored; IRT fallback). Table S10 is the full L1–L7 audit by dataset, split, and unique partition. They are not printed in the main PDF. Locked ASSISTments T-KT ECE 0.1136/0.2280 are verified, not retrained, by `scripts/rebuild_locked_tables.sh`.

Table 1: Four-unique-partition event-level calibration by train-only frequency stratum (same aggregation and $M = 15$ equal-width bins as main-text Table 5). N is the mean test-event count across four unique learner partitions. Flags: Reliable (R) $N \geq 1000$; Limited (L) $100 \leq N < 1000$. Junyi sparse is empty. T-KT is the local Transformer KT baseline. Brier $=$ UNC $-$ RES $+$ REL is the binned decomposition already used in the manuscript; components need not sum exactly to Brier. IRT RES $= 0$ on these learner-based strata. Junyi T-KT dense N (3,232,614) differs from IRT/DKT (3,226,541) because those models were scored on separately processed test files; ASSISTments and XES3G5M share N across models.

Dataset	Model	Stratum	N	Flag	ECE	Brier	UNC	REL	RES
ASSISTments 2012	IRT	dense	523,971	R	0.0031 ± 0.0006	0.2103 ± 0.0003	0.2103 ± 0.0003	0.0000 ± 0.0000	0.0000
ASSISTments 2012	IRT	medium	5,963	R	0.1598 ± 0.0190	0.2742 ± 0.0075	0.2484 ± 0.0013	0.0258 ± 0.0062	0.0000
ASSISTments 2012	IRT	sparse	415	L	0.0777 ± 0.0555	0.2420 ± 0.0218	0.2337 ± 0.0129	0.0084 ± 0.0090	0.0000
ASSISTments 2012	DKT	dense	523,971	R	0.0602 ± 0.0022	0.1931 ± 0.0010	0.2103 ± 0.0003	0.0053 ± 0.0005	0.0223 ± 0.0004
ASSISTments 2012	DKT	medium	5,963	R	0.1621 ± 0.0256	0.2297 ± 0.0144	0.2484 ± 0.0013	0.0301 ± 0.0086	0.0479 ± 0.0053
ASSISTments 2012	DKT	sparse	415	L	0.2333 ± 0.0084	0.2502 ± 0.0098	0.2337 ± 0.0129	0.0624 ± 0.0038	0.0451 ± 0.0037
ASSISTments 2012	T-KT	dense	523,971	R	0.1136 ± 0.0066	0.2116 ± 0.0019	0.2103 ± 0.0003	0.0201 ± 0.0015	0.0186 ± 0.0004
ASSISTments 2012	T-KT	medium	5,963	R	0.1541 ± 0.0051	0.2272 ± 0.0086	0.2484 ± 0.0013	0.0273 ± 0.0021	0.0478 ± 0.0053
ASSISTments 2012	T-KT	sparse	415	L	0.2280 ± 0.0197	0.2525 ± 0.0116	0.2337 ± 0.0129	0.0594 ± 0.0104	0.0392 ± 0.0088
Junyi Academy	IRT	dense	3,226,541	R	0.0026 ± 0.0018	0.2078 ± 0.0007	0.2078 ± 0.0007	0.0000 ± 0.0000	0.0000
Junyi Academy	IRT	medium	3,706	R	0.1209 ± 0.0082	0.2579 ± 0.0033	0.2432 ± 0.0013	0.0147 ± 0.0021	0.0000
Junyi Academy	DKT	dense	3,226,541	R	0.0389 ± 0.0022	0.1804 ± 0.0008	0.2078 ± 0.0007	0.0022 ± 0.0002	0.0294 ± 0.0002
Junyi Academy	DKT	medium	3,706	R	0.2513 ± 0.0217	0.3077 ± 0.0144	0.2432 ± 0.0013	0.0741 ± 0.0123	0.0091 ± 0.0033
Junyi Academy	T-KT	dense	3,232,614	R	0.0792 ± 0.0051	0.1884 ± 0.0020	0.2079 ± 0.0003	0.0084 ± 0.0015	0.0279 ± 0.0008
Junyi Academy	T-KT	medium	3,836	R	0.1073 ± 0.0156	0.2307 ± 0.0103	0.2417 ± 0.0038	0.0149 ± 0.0046	0.0260 ± 0.0058
XES3G5M	IRT	dense	1,269,345	R	0.0025 ± 0.0022	0.1622 ± 0.0018	0.1622 ± 0.0018	0.0000 ± 0.0000	0.0000
XES3G5M	IRT	medium	12,889	R	0.0122 ± 0.0062	0.1696 ± 0.0034	0.1695 ± 0.0032	0.0002 ± 0.0001	0.0000
XES3G5M	IRT	sparse	1,969	R	0.0424 ± 0.0097	0.1875 ± 0.0055	0.1856 ± 0.0048	0.0019 ± 0.0008	0.0000
XES3G5M	DKT	dense	1,269,345	R	0.0378 ± 0.0005	0.1223 ± 0.0009	0.1622 ± 0.0018	0.0027 ± 0.0002	0.0420 ± 0.0009
XES3G5M	DKT	medium	12,889	R	0.0788 ± 0.0013	0.1186 ± 0.0025	0.1695 ± 0.0032	0.0102 ± 0.0008	0.0601 ± 0.0016
XES3G5M	DKT	sparse	1,969	R	0.1139 ± 0.0054	0.1391 ± 0.0063	0.1856 ± 0.0048	0.0180 ± 0.0011	0.0632 ± 0.0040
XES3G5M	T-KT	dense	1,269,345	R	0.1176 ± 0.0014	0.1562 ± 0.0018	0.1622 ± 0.0018	0.0184 ± 0.0005	0.0236 ± 0.0006
XES3G5M	T-KT	medium	12,889	R	0.1129 ± 0.0061	0.1366 ± 0.0051	0.1695 ± 0.0032	0.0176 ± 0.0018	0.0493 ± 0.0010
XES3G5M	T-KT	sparse	1,969	R	0.1254 ± 0.0047	0.1493 ± 0.0043	0.1856 ± 0.0048	0.0216 ± 0.0023	0.0567 ± 0.0018

Table 2: Complete within-KC controlled sparsification (seed 42; 30 originally dense KCs per dataset; learner-based fold 0). Δ = reduced−full. Positive ΔECE means worse calibration after reducing training rows for the same KC. 95% CIs are bootstrap over KCs. T-KT is the local Transformer KT baseline, not published SimpleKT. This table reports every dataset × model × $\{500, 100, 50\}$ cell; none is omitted. Not a causal law for real-world sparsity.

Dataset	Model	Rows	ΔECE [95% CI]	ΔREL [95% CI]	Share ECE↑	n_{KC}
ASSISTments 2012	DKT	500	−0.047 [−0.060, −0.033]	−0.015 [−0.022, −0.010]	10%	30
ASSISTments 2012	DKT	100	−0.024 [−0.046, −0.001]	−0.008 [−0.017, +0.001]	30%	30
ASSISTments 2012	DKT	50	−0.017 [−0.034, +0.002]	−0.008 [−0.016, −0.002]	43%	30
ASSISTments 2012	T-KT	500	−0.026 [−0.047, −0.002]	−0.008 [−0.017, +0.003]	20%	30
ASSISTments 2012	T-KT	100	+0.002 [−0.023, +0.029]	0.000 [−0.013, +0.014]	43%	30
ASSISTments 2012	T-KT	50	+0.002 [−0.021, +0.025]	+0.001 [−0.011, +0.012]	47%	30
Junyi Academy	DKT	500	−0.021 [−0.041, −0.001]	−0.007 [−0.013, −0.002]	30%	30
Junyi Academy	DKT	100	−0.010 [−0.031, +0.011]	−0.003 [−0.009, +0.002]	37%	30
Junyi Academy	DKT	50	−0.004 [−0.022, +0.013]	−0.002 [−0.008, +0.003]	57%	30
Junyi Academy	T-KT	500	+0.101 [+0.081, +0.120]	+0.048 [+0.035, +0.060]	93%	30
Junyi Academy	T-KT	100	+0.132 [+0.109, +0.155]	+0.061 [+0.047, +0.077]	100%	30
Junyi Academy	T-KT	50	+0.135 [+0.110, +0.161]	+0.061 [+0.047, +0.074]	100%	30
XES3G5M	DKT	500	+0.142 [+0.104, +0.189]	+0.065 [+0.039, +0.103]	97%	30
XES3G5M	DKT	100	+0.137 [+0.107, +0.170]	+0.059 [+0.040, +0.081]	100%	30
XES3G5M	DKT	50	+0.161 [+0.127, +0.196]	+0.077 [+0.051, +0.106]	100%	30
XES3G5M	T-KT	500	+0.041 [+0.020, +0.063]	+0.021 [+0.010, +0.032]	83%	30
XES3G5M	T-KT	100	+0.083 [+0.061, +0.111]	+0.044 [+0.031, +0.058]	100%	30
XES3G5M	T-KT	50	+0.110 [+0.071, +0.156]	+0.070 [+0.040, +0.106]	93%	30

Table 3: Locked τ grid on ASSISTments 2012, learner-based fold 0 (seed 42), T-KT only. FAR $= P(y = 0 \mid p \geq \tau)$. Sparse occupancy is Limited ($N = 444$). Display threshold in the main text is $\tau = 0.7$ (locked FAR 0.196 / 0.268). The grid is not a search over sparse error.

τ	Dense FAR	Sparse FAR	ΔFAR	Sparse N_{advance}	Sparse flag
0.5	0.234	0.287	0.053	275	L
0.6	0.215	0.275	0.060	244	L
0.7	0.196	0.268	0.072	235	L
0.8	0.177	0.261	0.084	218	L

Table 4: Simulated practice-gate policies on frozen T-KT probabilities, ASSISTments 2012 learner-based fold 0 (seed 42). Same p as Table S5 / Table S3; CSV model name `simplekt` is the local T-KT baseline, not published SimpleKT. Policy A: global $\tau = 0.7$. Policy B: advance only if occupancy is Reliable (dense+medium). Policy C: $\tau = 0.8$ on the sparse stratum and $\tau = 0.7$ elsewhere. Population FAR/Miss use all test events (N = 534,150). Sparse FAR is undefined when $N_{\text{advance}} = 0$. Not a classroom trial.

Policy	Pop. N_{advance}	Pop. FAR	Pop. Miss	Sparse N_{advance}	Sparse FAR	Sparse Miss
A global $\tau = 0.7$	287,164	0.197	0.350	235	0.268	0.320
B Reliable-only	286,921	0.197	0.350	0	—	0
C sparse $\tau = 0.8$	287,147	0.197	0.350	218	0.261	0.289

Table 5: Simulated gate at $\tau = 0.7$, ASSISTments 2012 fold 0 (seed 42); T-KT and DKT only. Formerly main-text Table 5. Sparse occupancy is Limited. FAR 95% CIs: KC-cluster percentile, $B = 2000$. Locked cells: T-KT FAR 0.196 / 0.268. Not a classroom trial.

Model	Stratum	N	N_{advance}	$N_{\text{incorrect}}$	FAR [95% CI]	E[FAR]	Excess FAR	Miss
T-KT	dense	528,018	284,326	158,623	0.196 [0.186–0.208]	0.113	0.083	0.352
T-KT	sparse	444	235	197	0.268 [0.202–0.337]	0.050	0.218	0.320
DKT	dense	528,018	315,650	158,623	0.200 [0.190–0.211]	0.147	0.053	0.398
DKT	sparse	444	243	197	0.296 [0.221–0.383]	0.057	0.239	0.365

Seed-42 ΔFAR 95% CI (KC-cluster, $B = 2000$): T-KT $[0.006, 0.138]$; DKT $[0.019, 0.175]$.

Table 6: Gate robustness at $\tau = 0.7$ on ASSISTments 2012 (primary unit: four unique learner partitions). Formerly main-text Table 6. T-KT Mean ΔFAR is the partition-level mean 0.056 (the SD column is the partition range 0.015–0.087) after averaging seeds 2025 and 2026 first. The five training runs have mean 0.047 (sd 0.033) because those two seeds share one split. DKT remains a five-run summary. Mean N, N_{advance}, and $N_{\text{incorrect}}$ are sparse-stratum denominators.

Model	Mean ΔFAR	SD	Runs ΔFAR> 0	Mean N	Mean N_{advance}	Mean $N_{\text{incorrect}}$
T-KT	0.056	0.015–0.087	4/4 unique partitions	413	227	155
DKT	0.033	0.048	3/5 runs	413	226	155

Table 7: Three train-only KC-frequency cut grids on ASSISTments 2012 (T-KT and DKT). Four-partition means (seeds 2025/2026 averaged first). Dense: $f \geq t_2$. Sparse: $t_0 \leq f < t_1$. Main reprints Table 5 ECE (T-KT 0.1136/0.2280; DKT 0.0602/0.2333). Alt grids use the same 15-bin ECE on the frozen rerun CSVs. Official SimpleKT is not regrouped here. Not a new model.

Model	Cut grid ($t_0/t_1/t_2$)	Dense ECE	N dense	Sparse ECE	N sparse	ΔECE
T-KT	Main (20/100/500) (Table 5)	0.1136	523,971	0.2280	415	0.1144
T-KT	Alt-1 (10/50/250)	0.1138	528,522	0.2397	303	0.1260
T-KT	Alt-2 (30/150/750)	0.1135	522,485	0.2184	685	0.1049
DKT	Main (20/100/500) (Table 5)	0.0602	523,971	0.2333	415	0.1731
DKT	Alt-1 (10/50/250)	0.0610	528,522	0.2463	303	0.1853
DKT	Alt-2 (30/150/750)	0.0600	522,485	0.2087	685	0.1487

Table 8: Four-partition AUC and ACC by train-only frequency stratum (seeds 2025/2026 averaged first). Complements main-text Table 4 (overall AUC/ACC), Table 5 (ECE), and Table 6 (Brier). ASSISTments and Junyi: `analysis/summary_4part_bucket.csv`. XES3G5M: masked a2b summary (same N as Table 5). T-KT is the local Transformer, not published SimpleKT. Official SimpleKT [4] ASSISTments AUC is the four-partition mean from `p0_official_simplekt_assist_ece.json` (ACC per stratum not exported). Junyi sparse is empty. Does not change locked T-KT ECE 0.1136/0.2280.

Dataset	Model	Stratum	N	AUC	ACC
ASSISTments 2012	IRT	dense	523,971	0.5000 ± 0.0000	0.6993 ± 0.0007
ASSISTments 2012	IRT	medium	5,963	0.5000 ± 0.0000	0.5363 ± 0.0190
ASSISTments 2012	IRT	sparse	415	0.5000 ± 0.0000	0.6184 ± 0.0555
ASSISTments 2012	DKT	dense	523,971	0.6968 ± 0.0014	0.7185 ± 0.0014
ASSISTments 2012	DKT	medium	5,963	0.7574 ± 0.0141	0.6897 ± 0.0162
ASSISTments 2012	DKT	sparse	415	0.7287 ± 0.0106	0.6978 ± 0.0224
ASSISTments 2012	T-KT	dense	523,971	0.6822 ± 0.0024	0.6997 ± 0.0033
ASSISTments 2012	T-KT	medium	5,963	0.7546 ± 0.0153	0.6912 ± 0.0130
ASSISTments 2012	T-KT	sparse	415	0.7167 ± 0.0225	0.6962 ± 0.0108
Junyi Academy	IRT	dense	3,226,541	0.5000 ± 0.0000	0.7055 ± 0.0018
Junyi Academy	IRT	medium	3,706	0.5000 ± 0.0000	0.5820 ± 0.0083
Junyi Academy	DKT	dense	3,226,541	0.7322 ± 0.0010	0.7345 ± 0.0015
Junyi Academy	DKT	medium	3,706	0.6054 ± 0.0299	0.5839 ± 0.0172
Junyi Academy	T-KT	dense	3,232,614	0.7231 ± 0.0030	0.7275 ± 0.0015
Junyi Academy	T-KT	medium	3,836	0.6865 ± 0.0241	0.6442 ± 0.0191
XES3G5M	IRT	dense	1,269,345	0.5000 ± 0.0000	0.7962 ± 0.0031
XES3G5M	IRT	medium	12,889	0.5000 ± 0.0000	0.7838 ± 0.0057
XES3G5M	IRT	sparse	1,969	0.5000 ± 0.0000	0.7536 ± 0.0093
XES3G5M	DKT	dense	1,269,345	0.8177 ± 0.0009	0.8320 ± 0.0015
XES3G5M	DKT	medium	12,889	0.8608 ± 0.0030	0.8434 ± 0.0056
XES3G5M	DKT	sparse	1,969	0.8579 ± 0.0087	0.8253 ± 0.0096
XES3G5M	T-KT	dense	1,269,345	0.7525 ± 0.0010	0.8054 ± 0.0029
XES3G5M	T-KT	medium	12,889	0.8337 ± 0.0022	0.8333 ± 0.0065
XES3G5M	T-KT	sparse	1,969	0.8466 ± 0.0094	0.8156 ± 0.0068
ASSISTments 2012	SimpleKT [4]	dense	523,971	0.7687	—
ASSISTments 2012	SimpleKT [4]	medium	5,963	0.8238	—
ASSISTments 2012	SimpleKT [4]	sparse	415	0.7712	—

Table 9: Baseline inventory against Survey Roadmap v3.0 §1.2 / item 6. P0 is a diagnostic paper: baselines illustrate the protocol. BKT is not scored. IRT fills the classical slot after pyBKT 1.4.1 degenerated on ASSISTments seed 42 (roadmap fallback: two stable baselines plus a documented classical reference). Official SimpleKT [4] is the pyKT-family strong baseline on ASSISTments only; [4] does not report ASSISTments 2012, so this paper does not claim a 2–3% match to a published pyKT cell on that log. T-KT is a local Transformer and is not [4].

Roadmap slot	Required level	Scored in this P0	Note
BKT	Minimum (classical)	No	pyBKT degenerate; IRT used inste
IRT	Fallback classical	Yes (3 logs)	Base-rate / Rasch reference
DKT	Minimum	Yes (3 logs)	Local sequence baseline
simpleKT or AKT	Minimum (strong)	Official SimpleKT on ASSISTments	Not a published 2012 pyKT cell
T-KT (local)	Diagnostic Transformer	Yes (3 logs)	Not published SimpleKT [4]
AKT / DKVMN / sparseKT	Good / Excellent	No	Out of P0 scope

Table 10: Full L1–L7 leakage audit on frozen split files (same train-only rules as main-text Table 3). Learner folds 0, 1, 2, 4 are the four unique student partitions (seeds 42, 2024, 2025, 2027). Fold 3 is omitted because it shares fold 2 (seeds 2025/2026). Temporal is the seed-42 complementary split only. L2, L3, L5, and L6 are the shared pipeline (train-only transforms, static platform tag, no test-fit calibration map, final checkpoint). L4 empty = no KC with $20 \leq f_{\text{train}} < 100$. L7 empty / I = no $f = 0$ test events, or $N < 100$. Table 3 Assist L7 is the compact fold-0 label PASS (I); the fold-0 split file has no $f = 0$ test events and fold 2 has $N = 4$. XES3G5M drops $kc\_id < 0$. Not a new forensic method.

Dataset	Split	Fold	L1	L2	L3	L4	L5	L6	L7
ASSISTments	learner	0	PASS	PASS	PASS	PASS	PASS	PASS	PASS (empty)
ASSISTments	learner	1	PASS	PASS	PASS	PASS	PASS	PASS	PASS (empty)
ASSISTments	learner	2	PASS	PASS	PASS	PASS	PASS	PASS	PASS (I)
ASSISTments	learner	4	PASS	PASS	PASS	PASS	PASS	PASS	PASS (empty)
ASSISTments	temporal	0	PASS	PASS	PASS	PASS	PASS	PASS	PASS
Junyi	learner	0	PASS	PASS	PASS	PASS (empty)	PASS	PASS	PASS (empty)
Junyi	learner	1	PASS	PASS	PASS	PASS (empty)	PASS	PASS	PASS (empty)
Junyi	learner	2	PASS	PASS	PASS	PASS (empty)	PASS	PASS	PASS (empty)
Junyi	learner	4	PASS	PASS	PASS	PASS (empty)	PASS	PASS	PASS (empty)
Junyi	temporal	0	PASS	PASS	PASS	PASS	PASS	PASS	PASS
XES3G5M	learner	0	PASS	PASS	PASS	PASS	PASS	PASS	PASS (I)
XES3G5M	learner	1	PASS	PASS	PASS	PASS	PASS	PASS	PASS (I)
XES3G5M	learner	2	PASS	PASS	PASS	PASS	PASS	PASS	PASS (I)
XES3G5M	learner	4	PASS	PASS	PASS	PASS	PASS	PASS	PASS (I)
XES3G5M	temporal	0	PASS	PASS	PASS	PASS	PASS	PASS	PASS

Table 11: Cluster-aware T-KT ECE regression (Section IV.E). ASSISTments and Junyi rows are KC-fold observations at fold 0 with cluster-robust SE at $\mathrm{kc\_id}$. XES3G5M rows are the A2B-masked weighted fit ($n = 829$ unique KCs; padding excluded). Predictors are standardized within dataset. Not a causal frequency effect.

Dataset	Covariate	n_{obs}	Unique KCs	$\hat{\beta}$	SE_{cl}	95% CI	p
Primary: weighted WLS, cluster-robust SE							
ASSISTments 2012	$\log(1 + f_{\text{train}})$	478	261	-0.068	0.009	$[-0.086, -0.050]$	< 0.001
	Difficulty proxy	478	261	$+0.026$	0.005	$[+0.016, +0.036]$	< 0.001
	Item support	478	261	-0.006	0.003	$[-0.011, -0.000]$	0.046
	Learner exposure	478	261	$+0.016$	0.006	$[+0.004, +0.027]$	0.007
	Curriculum position	478	261	-0.005	0.004	$[-0.012, +0.003]$	0.217
Junyi Academy	$\log(1 + f_{\text{train}})$	2,645	1,326	-0.014	0.002	$[-0.018, -0.010]$	< 0.001
	Difficulty proxy	2,645	1,326	$+0.003$	0.001	$[+0.001, +0.005]$	0.008
	Item support	2,645	1,326	-0.001	0.001	$[-0.002, +0.001]$	0.280
	Learner exposure	2,645	1,326	$+0.001$	0.001	$[-0.002, +0.003]$	0.573
	Curriculum position	2,645	1,326	-0.004	0.001	$[-0.005, -0.002]$	< 0.001
XES3G5M	$\log(1 + f_{\text{train}})$	829	829	-0.028	0.007	$[-0.042, -0.014]$	< 0.001
	Difficulty proxy	829	829	$+0.080$	0.004	$[+0.072, +0.088]$	< 0.001
	Item support	829	829	-0.007	0.001	$[-0.010, -0.005]$	< 0.001
	Learner exposure	829	829	$+0.013$	0.004	$[+0.005, +0.021]$	0.001
	Curriculum position	829	829	-0.014	0.002	$[-0.019, -0.009]$	< 0.001
Sensitivity: unweighted OLS, same cluster-robust SE							
ASSISTments 2012	$\log(1 + f_{\text{train}})$	478	261	-0.060	0.014	$[-0.086, -0.033]$	< 0.001
	Difficulty proxy	478	261	$+0.029$	0.010	$[+0.009, +0.049]$	0.004
	Item support	478	261	-0.012	0.003	$[-0.019, -0.005]$	< 0.001
	Learner exposure	478	261	$+0.021$	0.007	$[+0.007, +0.036]$	0.005
	Curriculum position	478	261	-0.009	0.007	$[-0.022, +0.004]$	0.170
Junyi Academy	$\log(1 + f_{\text{train}})$	2,645	1,326	-0.022	0.002	$[-0.026, -0.018]$	< 0.001
	Difficulty proxy	2,645	1,326	$+0.005$	0.001	$[+0.004, +0.007]$	< 0.001
	Item support	2,645	1,326	-0.000	0.001	$[-0.002, +0.001]$	0.728
	Learner exposure	2,645	1,326	$+0.009$	0.002	$[+0.004, +0.013]$	< 0.001
	Curriculum position	2,645	1,326	-0.001	0.001	$[-0.003, +0.000]$	0.106
XES3G5M	$\log(1 + f_{\text{train}})$	829	829	-0.008	0.012	$[-0.031, +0.015]$	0.488
	Difficulty proxy	829	829	$+0.070$	0.006	$[+0.057, +0.083]$	< 0.001
	Item support	829	829	-0.009	0.002	$[-0.014, -0.004]$	< 0.001
	Learner exposure	829	829	$+0.007$	0.008	$[-0.009, +0.023]$	0.396
	Curriculum position	829	829	-0.017	0.004	$[-0.025, -0.009]$	< 0.001